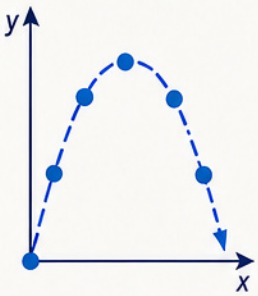
PHY4605 COURSE GUIDE

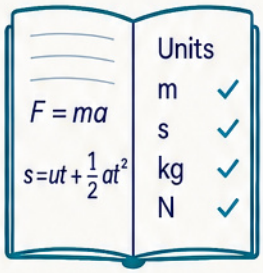
How to use the course materials

1 DURING CLASS
LECTURE DECK



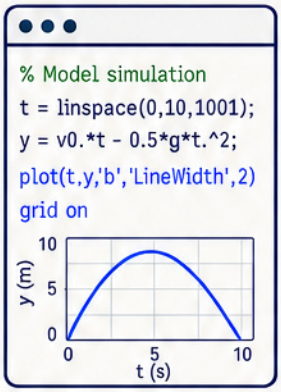
Preview the physical question, algorithm, evidence, and interpretation.

2 BEFORE CLASS
LEARNING NOTE



Read the model, units, worked explanation, and method boundary.

3 DURING CLASS
MATLAB LIVE SCRIPT



Run from a fresh session; trace, modify, validate, and interpret.

4 DURING PRACTICAL
PRACTICAL PACKET



Work in a group; record AI decisions; submit evidence; prepare for defence.

1 BEFORE CLASS → **2 DURING CLASS** → **3 PRACTICAL** → **4 AFTER CLASS**

Predict → model → units → algorithm → code → validation → interpretation

GOOGLE CLASSROOM = official record

MATLAB = computational evidence

GEMINI NOTEBOOK = Socratic tutor

Written tests and final examination are AI-free.

PHY4605 COURSE GUIDE

14-week roadmap and assessment windows

14-WEEK ROADMAP

W01	Arrays and Plots
W02	Numerical Algorithms
W03	Linear Systems
W04	Parameter Sweeps
W05	Root Finding
W06	Numerical Differentiation
W07	Numerical Integration
W08	Ordinary Differential Equations
W09	Data Fitting
W10	Monte Carlo Sampling
W11	Sensitivity Analysis
W12	Method Selection
W13	Capstone Presentation
W14	Capstone Presentation (continued)

ASSESSMENT WINDOWS

TEST 1

UPM Assessment I: Weeks 4–6

26 Oct–08 Nov 2026 and 16–22 Nov 2026

TEST 2

UPM Assessment II: Weeks 8–12

30 Nov 2026–03 Jan 2027

FINAL EXAMINATION

21 Jan–07 Feb 2027

Revision break: 18–20 Jan 2027

CAPSTONE PATHWAY

1. **W03** — Problem-space introduction
2. **W04** — Preferred and alternate choice
3. **W06** — Feasibility proposal
4. **W07** — Model, method, and validation approval
5. **W08–W12** — Process checkpoints and capstone studio
6. **W13–W14** — Presentation, defence, and reproducibility handoff

W13–W14: capstone presentation, defence, and handoff.

Planning windows follow the UPM Bachelor Academic Calendar 2026/2027 Semester 1.